General weights: uniform quadratic mixing

Result. Lean proves the qualified uniform-support Theorem2 of Hildebrand1994: for every fixed finite $k \geq 2$ and strictly positive normalized weight vector, a positive γ depending only on weights and ε works over every prime modulus and every distinct support. At all integer times $t \geq \gamma p^2$, every event differs from uniform by less than ε. The constant is chosen before the modulus/support. [pcc:unverified]

Actual arbitrary weights	Normalized label PMF; product law = two iid binds Equal-mass label-pair swap preserves both margins
Exact scaled drift	Normalized difference; $V = k(p - k)$, $V \leq p^2/4$ Expected drop $2\, w_a\, w_b$ until agreement
Qualified source result	Event distance $\leq p^2/(8\, w_a\, w_b\, t)$ All supports; γ depends only on fixed weights and ε

Actual probability and coupling. A normalized NNReal vector constructs an actual Mathlib PMF, bound afresh for each increment. The actual product-label PMF equals two independent binds. A concrete transposition on the two selected ordered pairs preserves product mass and both marginal expectations. After agreement, increments are shared. The difference divided by the selected increment difference moves± 1 with probability $w_a \cdot w_b$ each. Actual modular potential drift, coalescence, finite expectation identities, event inequalities and uniform stationarity are proved, not premises. [pcc:unverified]

Explicit bound and scope. Two positive weights at distinct increments suffice for $p^2/(8 w_a w_b t)$; $\gamma = 1/(4 w_a w_b \varepsilon)$ gives distance$\leq \varepsilon/2 < \varepsilon$. The source-shaped wrapper retains all positive weights and support injectivity. The product coupling gives $p^2/(2t)$ at two equal halfweights; the prior antithetic packet retains its sharper $p^2/(4t)$. This component extends scope, with no sharper-constant claim. [pcc:unverified]

Direct prime-deletion counter. Over $Z/4$, actual distinct increments0/2 with both weights1/2 stay in the actual two-element event$\{0,2\}$. Event probability1 versus uniform1/2 holds at every time. Lean directly rules out any positive quadratic time constant giving every event distance$< 1/4$. Sourcep192 already warns about composite-modulus subgroup confinement; no novelty claim. [pcc:unverified]

Primary review. Hildebrand1994, PTRF100,191–203: all13 published pages/proofs/discussion/references read and all13 rendered pages visually inspected; Theorem2p193/proof199–200, definitions/warningp192. Established coupling/potential method credited to Levin–Peres, contributions Wilmer, selected Proposition4.7/Theorem5.4/Corollary5.5/cycle5.3.2/Example10.27 context. Full book not read. Paper event normalization used; no identification with differently normalized Mathlib tvDist. [pcc:unverified]

Checks and remainder. Wholegate59/0failures/0trusted; all102 build/processing/provenance files retained in35objects, including every failed state and compiler-panic log. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified. Stronger random-support Theorem1 exponent, static rho and collision constants remain open. KB/Conductor unavailable, bounded fallback before code. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]